enough to learn their predictions, and we had no way to observe geometrodynamics in the astronomical universe.

I've devoted much of my career to changing this. I cofounded LIGO with the goal of observing geometrodynamics in the distant universe. In 2000, when I turned my LIGO roles over to others, I cofounded a research group at Caltech aimed at simulating geometrodynamics on supercomputers, by solving Einstein's relativistic equations numerically. We call this project SXS: Simulating eXtreme Spacetimes. It is a collaboration with Saul Teukolsky's research group at Cornell University, and others.

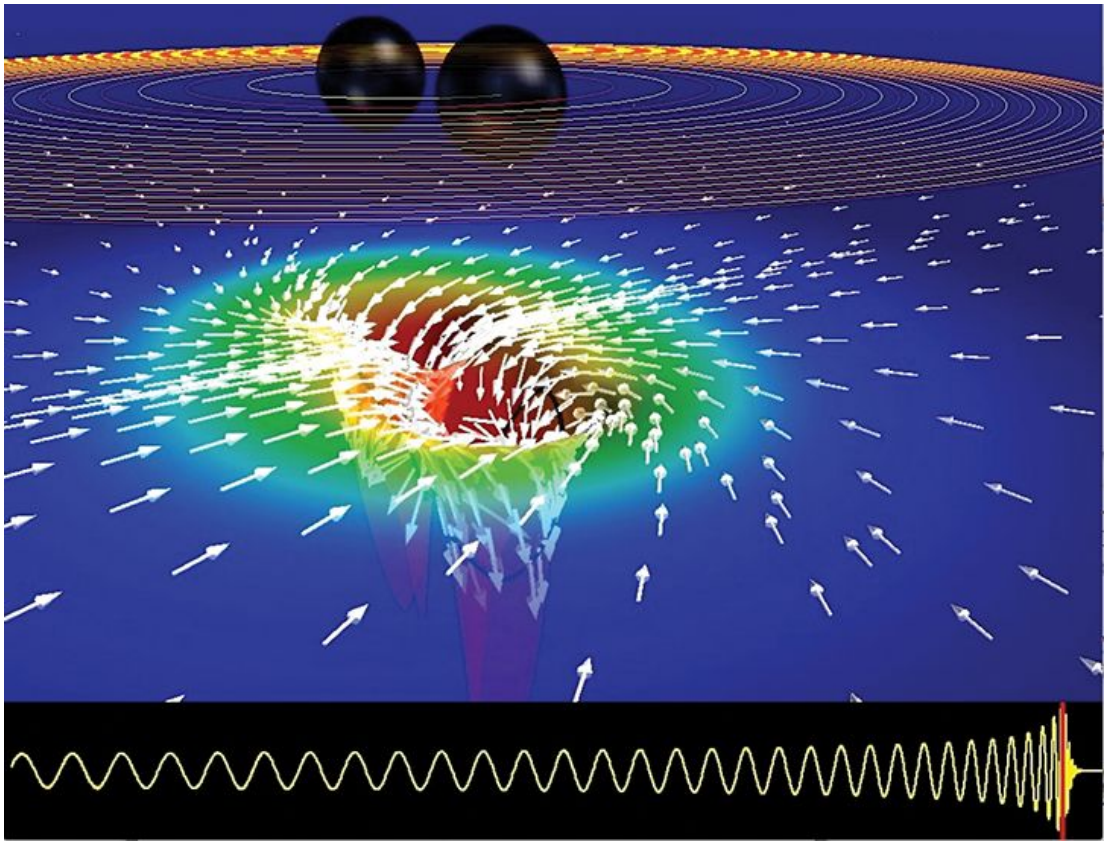


Fig. 16.9. Simulation of two black holes at their moment of collision. *Top*: The holes' orbits and shadows as seen in our universe. *Middle*: The holes' warped space and time as seen from the bulk, with arrows showing the dragging of space into motion and colors the warping of time. *Bottom*: The emitted gravitational waveform. This simulation is for identical, nonspinning black holes. *[From a movie by Harald Pfeifer of a simulation by the SXS team.]*